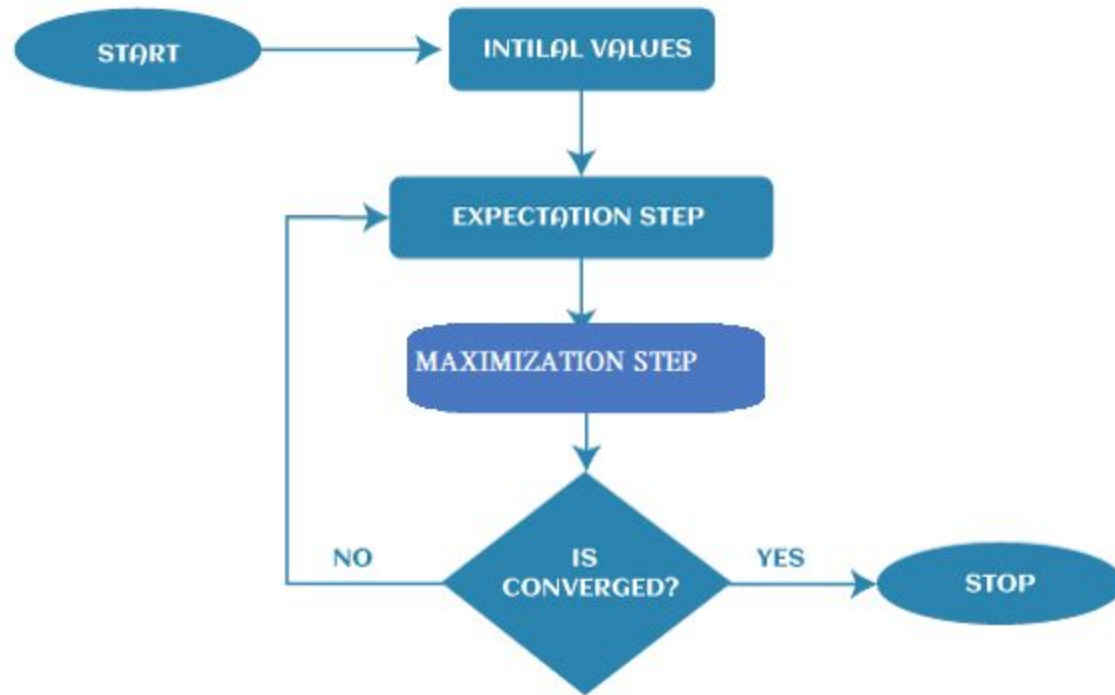


Expectation Maximization

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Expectation Maximization

- EM is maximum-likelihood estimation in the presence of latent/unobserved variables
- Iterative algorithm: Iterates these two steps until convergence
 - a. E-Step
 - b. M-Step
- EM is a numerical technique for maximum likelihood estimation
- Iterative algorithm
 - a. E-step (Expectation)
 - b. M-step (Maximization)
- The maximum likelihood of the data strictly increases with each subsequent iteration, thus it is guaranteed to reach a local maximum or saddle point
- It is often used in computer vision and NLP



[source](#)

Two Coin Flips Example

EM: General Approach to Many Problems

- EM is used to solve many different tasks with the same problem: calculating MLE is difficult due to an unobserved/latent variable
- Two coin flips is a classical example:
 - We have two coins: A and B
 - Each has some unknown probability of getting heads: θ_A and θ_B
 - Imagine each time, we randomly pick a coin and flip it ten times:

Flips
HTTTHHTHTH
HHHHTHHHHH
HTHHHHHTHH
HTHTTTTHHTT
THHHTHHHTH



EM: Two Coin Flips Example

- The problem is we don't know which coin each row comes from. We can solve this using the EM algorithm. The goal is to assign A or B to each row depending on which coin it comes from most probably

Coin	Flips
B	HTTTHHTHTH
A	HHHHTHHHHH
A	HTHHHHHTHH
B	HTHTTTHTTT
A	THHHTHHHTH



EM: Two Coin Flips Example - Initial Guess

- Let's assume that $\theta_A = 0.6$ and $\theta_B = 0.5$



EM: Two Coin Flips Example - E-step

- In this step, given the initial values for θ_A and θ_B (parameters are fixed), we want to update the allocation of heads/tails to latent variables (coin identity)
- We can use the binomial distribution formula to estimate the probability that we get a specific sequence of Head and Tail flips given each of our coins (n is the number of flips in a series and x is the number of heads):

$$P(E | Z_A) = P(\text{HHHHHHHHHT} | \text{A chosen}) = \binom{n}{x} \theta_A^x (1 - \theta_A)^{n-x}$$

$$P(E | Z_B) = P(\text{HHHHHHHHHT} | \text{B chosen}) = \binom{n}{x} \theta_B^x (1 - \theta_B)^{n-x}$$

- Now, We can use the Bayes Theorem and the law of total probability to estimate the probability that a certain coin was used to generate the flips:

$$P(Z_A | E) = \frac{P(E | Z_A) P(Z_A)}{P(E | Z_A) P(Z_A) + P(E | Z_B) P(Z_B)}$$

$$P(Z_B | E) = \frac{P(E | Z_B) P(Z_B)}{P(E | Z_A) P(Z_A) + P(E | Z_B) P(Z_B)}$$

EM: Two Coin Flips Example - E-step

- Then we can use these probabilities as weights and estimate the expected number of heads for each coin (**Expectation step**)

N heads	$P(Z_A E)$	$P(Z_B E)$	$Heads_A$	$Tails_A$	$Heads_B$	$Tails_B$
5	0.45	0.55	2.2	2.2	2.8	2.8
9	0.8	0.2	7.2	0.8	1.8	0.2
8	0.73	0.27	5.9	1.5	2.1	0.5
4	0.35	0.65	1.4	2.1	2.6	3.9
7	0.65	0.35	4.5	1.9	2.5	1.1
		<i>Total</i>	21.3	8.6	11.7	8.4

EM: Two Coin Flips Example - M-step

- In this step, we assume the allocations to latent variable (Z) are fixed and we update the parameters.
- Using this updated data of heads, we can update our guess for our parameters using normal maximum likelihood estimation(**Maximization step**):

$$\theta_A^1 = \frac{21.3}{21.3 + 8.6} = 0.71$$

$$\theta_B^1 = \frac{11.7}{11.7 + 8.4} = 0.58$$

- Now we will use the above as an input and repeat the E and M steps again. We can repeat this until convergence (likelihood does not increase anymore) or for a preset number of times such as 1000 iterations

EM: Two Coin Flips Example

- [Expectation-maximization-Coin flips- formulas and explanation](#)
- [Expectation Maximization with Coin Flips: Code](#)